基于 Zynq 的音频可视化与本地听歌识曲系统

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项目概览

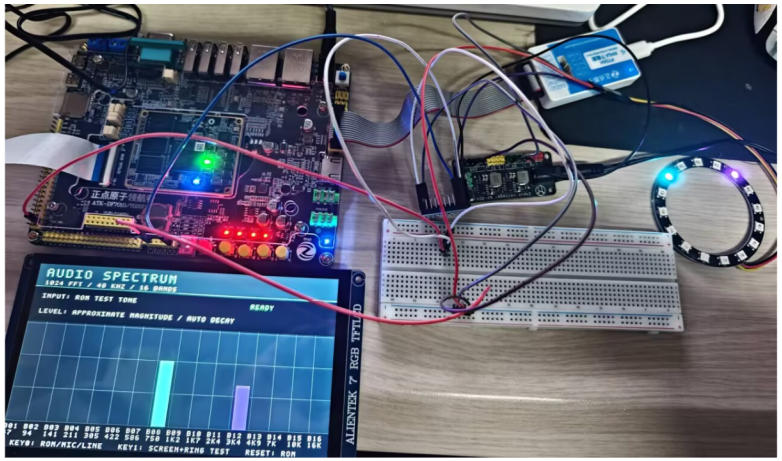


图1 开发板、LCD 与16灯灯环

板载 MIC 与 LINE IN 采集音频，FPGA 完成实时频谱与灯效。电脑匹配本地曲库，并将歌名和演唱者回显到板端 LCD。

表1 硬件与工具

FPGA	XC7Z010CLG400-1
采音	ES8388，48 kHz
屏幕	7英寸 RGB LCD，1024×600
灯环	16灯 WS2812B
工具	Vivado 2022.2

系统设计

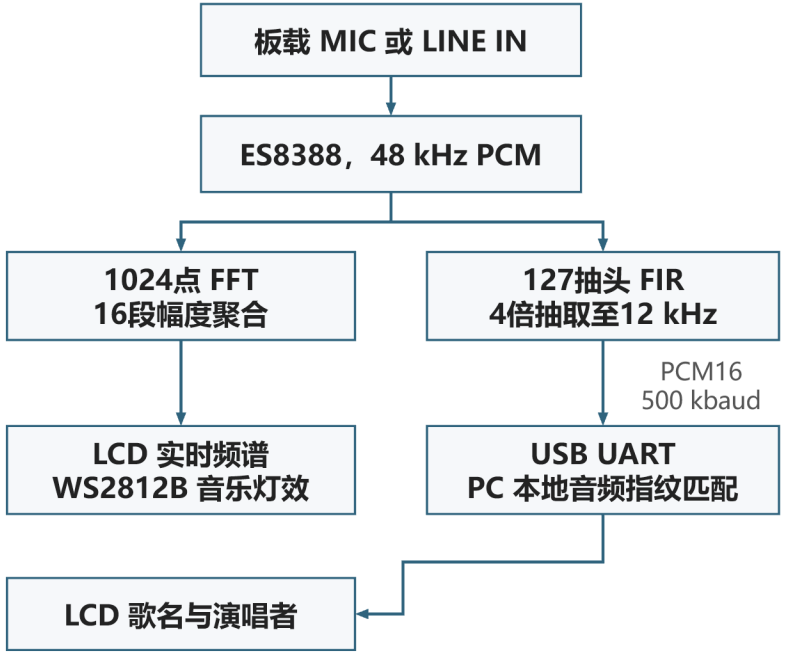


图2 并行频谱与识曲数据通路

算法与显示保护

频谱：1024点 FFT，频点间隔46.875 Hz。

识曲：Hann窗提取局部峰值并配对，按时间偏移一致性投票确认歌曲。

回显：连续证据确认，CRC16 校验与双缓冲保护文字更新。

测试与性能

表2 实现后资源占用

资源	使用	可用	占比
LUT	7,232	17,600	41.09%
FF	10,560	35,200	30.00%
BRAM	23	60	38.33%
DSP	7	80	8.75%

BRAM 按36 Kb块计，统计包含 ILA 调试核。

主时钟50 MHz，WNS +0.379 ns

WHS +0.025 ns，已约束路径时序通过。

表3 验证摘要

RTL 仿真	21项通过
PC 回归	33项通过
LINE IN 已知片段	5首×3片段，15/15
库外歌曲	3首均拒识
实物功能	MIC采集、灯环联动 与LCD歌名回显

本地曲库16首、11位歌手。仅识别已录入的原版录音，以上结果对应已测样本。

后续：扩充曲库，验证复杂噪声场景。

AMD

Audio Visualization and Local Song Recognition on Zynq

Team: Haoran Liu, Kaixin Mao, Hang Shi Supervisor: Dandan Wei

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SYSTEM OVERVIEW

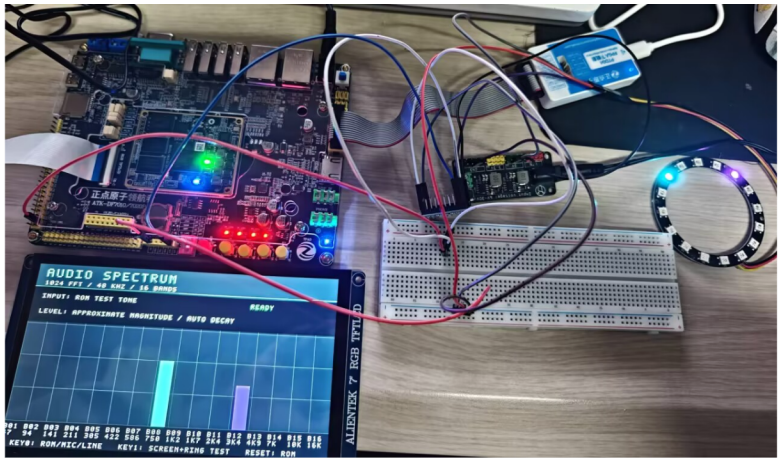


Fig. 1 Prototype with LCD and 16-LED ring

The on-board MIC or LINE IN feeds audio to the FPGA for spectrum and RGB lighting. A PC matches a local song library and sends the title and artist back to the LCD.

Table 1 Hardware and tools

FPGA	XC7Z010CLG400-1
Audio	ES8388, 48 kHz
Display	7-inch RGB LCD, 1024×600
RGB ring	16 WS2812B LEDs
Tools	Vivado 2022.2

SYSTEM DESIGN

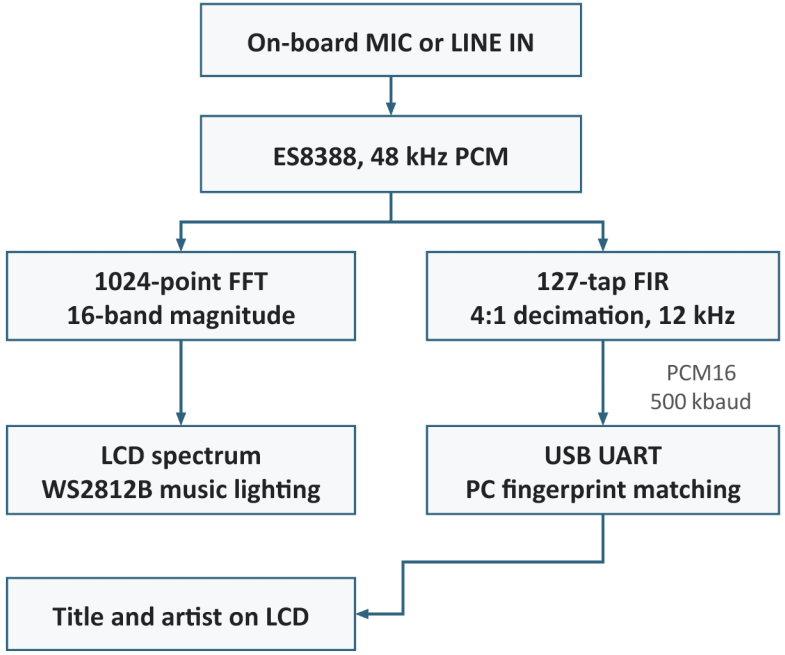


Fig. 2 Parallel spectrum and recognition paths

Algorithms and display integrity

Spectrum: 48 kHz / 1024 = 46.875 Hz per FFT bin.

Matching: Hann-window spectral peaks form paired fingerprints. Votes align in time.

Display: consistent evidence, CRC16 checks and double buffering protect text updates.

TESTS AND PERFORMANCE

Table 2 Post-route resource utilization

Resource	Used	Available	Usage
LUT	7,232	17,600	41.09%
FF	10,560	35,200	30.00%
BRAM	23	60	38.33%
DSP	7	80	8.75%

BRAM counts 36 Kb tiles. Figures include the ILA.

Main clock: 50 MHz. WNS: +0.379 ns.

WHS: +0.025 ns. Constrained paths meet timing.

Table 3 Validation summary

RTL simulation	21 cases passed
PC regression	33 tests passed
LINE IN samples	5 tracks × 3 excerpts, 15/15
Unknown tracks	3 tracks rejected
Hardware	MIC capture, RGB response and LCD song title feedback

Local library: 16 tracks by 11 artists. Recognition targets enrolled recordings. Results refer to tested samples.

Next: broader library and noise testing.

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